

Evidence and Sources

Project: Transparent Book Recommender

1. Evidence and Sources

Teach the learning cycle

Primary target: Explain the mechanism and terminology of claim support and citation evaluation

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-2/29, ai-3/16. Reuse a familiar case for recall or

modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Prompt Experiments

Without opening the old lesson, explain one key idea from Prompt Experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of controlled prompt experiments

Reference example, not the only acceptable wording: Teacher explanation: A prompt experiment compares complete specifications using the same task cases. Define success before inspecting outputs; change one intended instruction and preserve other settings. Evaluate both prompts on paired cases with fixed model/version, source records, output contract and scoring rubric. Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

Image and Voice Pipelines

Without opening the old lesson, explain one key idea from Image and Voice Pipelines. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of speech pipelines and multimodal evidence

Reference example, not the only acceptable wording: Teacher explanation: A multimodal pipeline may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly. Track source modality, segment/time references, transcription, interpretation and final claims separately. Compare conflicting evidence without assuming one modality always wins. Ask for clarification when the task needs unresolved information.

Recommendation Scores

Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a weighted score

Reference example, not the only acceptable wording: Score 8.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Two catalogue records list 96 and 112 pages for a title. A helper averages them and reports 104. Explain the mistake and a better verification process.

Concept to explain

Claims require sources matched to the question and checked independently.

Term · Meaning

Term: claim

Meaning: A statement that can be examined using evidence and reasoning.

Term: source

Meaning: The record, person or material from which information is obtained.

Term: independent

Meaning: Produced or assessed without the specific influence being excluded; state whether this means uncoached work or separate test data.

Term: corroborate

Meaning: Support a claim with additional relevant evidence, ideally independent of the first source.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Two 45-minute sessions. Citation evaluation checks source existence, exact passage support and coverage of answer claims. A real citation can still be irrelevant. Use the supplied constructed scenarios to make the mechanism inspectable. Collect W4 before sharing solutions. Students should name intermediate evidence and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the inputs, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from predictions or unmeasured conditions. If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

End goal for the pictorial case

By the end, I can separate citation presence, link validity and support for an exact claim.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. Claim: a book won an award in 2024.
2. Find the award organization's record and the publisher's page.
3. Check exact title, year, edition, and whether sources are independent.
4. If evidence conflicts or is absent, mark 'not verified' rather than guess.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

An answer says 80 pages, but the catalogue for that edition says 120.

What evidence guides correction?

Exact explanation to model

Check the exact edition and authoritative catalogue record. Record the supporting source.

Explain the added worked example

Citation evaluation checks source existence, exact passage support and coverage of answer claims. A real citation can still be irrelevant.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

For this classroom rubric count a claim as supported only when its cited passage entails it. Separately count correct citation links. Use the number of factual claims as the support denominator.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term · Meaning

Term: Entailment

Meaning: Whether the cited passage supports the specific claim

Term: Citation coverage

Meaning: Which claims include source references

Term: Link validity

Meaning: Whether a source reference resolves to the intended record

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

An answer has four factual claims. Three have citations, but only two are supported by their cited passages. All links exist.

Claim support is $2/4=50\%$; citation presence is $3/4=75\%$; link existence among three citations is $3/3$. These measure different failures.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Evidence and Sources

A fictional catalogue answer contains three factual statements and cites existing records.

The facts we will use

Record R: book Ocean Maps has 80 pages and is available on shelf 2.

Claims: 80 pages [R]; shelf 2 [R]; suitable for every child [R]. All three citations resolve to R.

AI ACADEMY / AI-2/29

Different citation checks

percent



Bar length shows the supplied value; the marked baseline is zero.

Evidence and Sources

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Break into claims

Count three factual claims rather than treating the answer as one blob.



2. Check links

All three citation links resolve to the intended R.



3. Check exact source support

R supports page count and shelf, not universal suitability.



4. Report separately

Citation presence 3/3; link validity 3/3; supported claims 2/3.

CHANGE ONE THING

Change the third claim to "the record does not assess suitability".

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Break into claims

What happens and why: Count three factual claims rather than treating the answer as one blob.

Step: 2. Check links

What happens and why: All three citation links resolve to the intended R.

Step: 3. Check exact source support

What happens and why: R supports page count and shelf, not universal suitability.

Step: 4. Report separately

What happens and why: Citation presence 3/3; link validity 3/3; supported claims 2/3.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

An existing citation is not automatically evidence for every nearby sentence. Read the cited passage and compare its exact scope with the claim. The universal suitability statement extends beyond what R says. Remove or qualify unsupported content; adding more copies of the same link does not create the missing evidence.

What this example does not establish

Source support is relative to supplied records; those records can themselves be incomplete, outdated or incorrect.

Change one thing in the pictured case

Change: Change the third claim to “the record does not assess suitability”.

Prediction: That statement accurately describes the supplied record’s limits.

Reason: R gives pages and shelf only; it contains no suitability assessment.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: An answer says 80 pages, but the catalogue for that edition says 120.

Write the question to explain: What evidence guides correction?

Reveal after predictions: Check the exact edition and authoritative catalogue record. Record the supporting

source.

Keep for an independent attempt: An answer cites a page about opening hours to support a claim about book prices. Is a citation enough? What must be checked?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: Claim support is $2/4=50\%$; citation presence is $3/4=75\%$; link existence among three citations is $3/3$. These measure different failures. Qualification: Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1: 0–7 minutes

Teacher action: Retrieve the prerequisite; identify the input, procedure and evidence source.

Stage: 7–18 minutes

Teacher action: Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.

Stage: 18–32 minutes

Teacher action: Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

Stage: 32–45 minutes

Teacher action: Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

Stage: Session 2: 0–8 minutes

Teacher action: Retrieve the worked reasoning and collect the guided response before discussing it.

Stage: 8–20 minutes

Teacher action: Compare a proposed boundary variant with the original case; identify what remains fixed.

Stage: 20–35 minutes

Teacher action: Collect W4 independently. Require the procedure trace and the bounded conclusion.

Stage: 35–45 minutes

Teacher action: Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Completion task: An answer has four factual claims. Three have citations, but only two are supported by their cited passages. All links exist. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Claim support is $2/4=50\%$; citation presence is $3/4=75\%$; link existence among three citations is $3/3$. These measure different failures.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Claim support and citation evaluation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more

demanding variation.

Pictorial practice question

Why is the original answer not fully supported despite having citations on every claim?

Reasoned teacher answer

One of the three claims is not entailed by its cited source; citation coverage and link validity do not measure claim support.

Changed-case reasoning: That statement accurately describes the supplied record's limits. R gives pages and shelf only; it contains no suitability assessment.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Highlight checkable claims in an answer, match each to the best source type, record evidence, and label supported / conflicting / unverified.

What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims.

A missing input may be replaced without recording the change: A replacement changes the evidence or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent evidence: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed

incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Check the exact edition and authoritative catalogue record. Record the supporting source.

Independent question

An answer cites a page about opening hours to support a claim about book prices. Is a citation enough? What must be checked?

Expected independent answer

No. Inspect whether the cited passage directly supports the price claim and is relevant and current for the task.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Any citation proves any nearby claim.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Identify ISBN or edition and consult its authoritative record or the supplied book. Report the supported page count for that edition; if unresolved, state uncertainty. Averaging conflicting records invents a value that may match no actual edition.

Explanations for the additional workbook examples

W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Teacher explanation: Citation evaluation checks source existence, exact passage support and coverage of answer claims. A real citation can still be irrelevant. For this classroom rubric count a claim as supported only when its cited passage entails it. Separately count correct citation links. Use the number of factual claims as the support denominator.

W2 Trace the worked case

An answer has four factual claims. Three have citations, but only two are supported by their cited passages. All links exist. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Claim support is $2/4=50\%$; citation presence is $3/4=75\%$; link existence among three citations is $3/3$. These measure different failures.

W3 Audit the limit

A report omits this qualification: Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims. State the observed result separately from the unverified or out-of-scope claim; preserve the original evidence rather than inventing a missing fact.

W4 Apply a new case independently

Five claims; four cited; three supported by cited passages; one cited ID is nonexistent. Compute support and citation presence and explain link limits.

Teacher explanation: Support= $3/5=60\%$; citation presence= $4/5=80\%$. Link checks reject the nonexistent ID; remaining real links still need entailment checks.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked input or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure. Require preserved settings, original outputs and data roles. Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims. The new case should test this particular mechanism rather than merely restate the original result.

Assess independent transfer

Selected case: Five claims; four cited; three supported by cited passages; one cited ID is nonexistent. Compute support and citation presence and explain link limits.

Selected case answer

Teacher explanation: Support= $3/5=60\%$; citation presence= $4/5=80\%$. Link checks reject the nonexistent ID; remaining real links still need entailment checks.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Transparent Book Recommender

Separate profile-based ranking evidence from factual book information and cite the latter.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record.

Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 30: Agents and Tools. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Support= $3/5=60\%$; citation presence= $4/5=80\%$. Link checks reject the nonexistent ID; remaining real links still need entailment checks. Require intermediate reasoning and the stated scope. Fresh retry: Six claims, five cited, four supported; all five links real. Expected reasoning: Support $4/6=2/3$; citation presence $5/6$; link existence $5/5$ does not verify every claim. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Abstention and clarification. Preserve the current case record and retrieve its evidence boundary before moving on.

Plan the delayed return

Suggested return points in this level: ai-2/30, ai-2/32, ai-2/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Attach source IDs to recommender explanation facts and separately audit whether each source supports its precise statement.